

SABARI MANI K

Embedded Systems • Power Electronics • Firmware & PCB Design

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ABOUT ME

Embedded Engineer with hands-on experience in firmware development, power electronics, PCB design, and hardware validation. Currently working at Fizix Solar Innovation, developing solar and battery-management systems involving protection circuits, voltage/current sensing, MOSFET and relay switching, and KiCad-based PCB design. Experienced in Embedded C, ARM microcontrollers, ESP32, RTOS, and communication protocols including UART, I²C, SPI, LoRa, GSM, and GPS. Built and tested embedded systems from circuit design and firmware development through hardware bring-up and debugging.

EDUCATION

2021 – 2025

Francis Xavier Engineering College

B.E. Electronics and Communication

CGPA: 8.11

2020 – 2021

Rosemary Matric Higher Secondary School

HSC

82.6%

EXPERIENCE

Oct 2025 – Present

**Fizix Solar Innovation
Pvt. Ltd., Chennai**

Embedded Electronics Engineer

- Designed and tested series/parallel battery switching circuits for solar power and battery-management systems.
- Developed voltage/current sensing and protection circuits using comparators, MOSFETs, and relays for under/over-voltage protection.
- Managed the PCB development cycle, including KiCad design, BOM preparation, assembly, bring-up, and hardware validation.
- Debugged hardware–software integration issues at the component level during system testing and validation.

May 2025 – Oct 2025

**Dr. MGR Educational
and Research Institute
– ARI, Chennai**

Embedded Electronics Engineer

- Developed embedded firmware for ARM microcontrollers using UART, I²C, and PWM peripherals.
- Developed RTOS-based modules for automation systems using concurrent tasks.
- Tested and debugged embedded prototypes across hardware and firmware.

Jun 2023 – Jul 2023

**XYMA Analytics Pvt. Ltd.
, Chennai**

Electronics Engineering Intern

- Built a DS18B20-based temperature data logger for sensor data acquisition and monitoring.
- Prototyped a predictive-maintenance system involving sensor data acquisition and onboard data analysis.

PROJECTS

48V Solar Power Data Logger | Fizix Solar Innovation

2026

- Designed a 48V power-monitoring data logger for measuring and recording voltage, current, power, and irradiance.
- Integrated INA238 for voltage/current monitoring, OPT4003 for light/irradiance measurement, and DS3231 RTC for time-stamped data acquisition.
- Designed the PCB and developed the embedded system for sensor acquisition, data logging, and system validation.

Tech: ESP32-S3, INA238, OPT4003, DS3231, I²C, KiCad, Embedded C

SolarDrive5 – Solar Power & USB-C Power System | Fizix Solar Innovation

2026

- Designed a solar-input power system with voltage monitoring, protection, and MOSFET-based power switching.
- Developed LM393 comparator-based voltage detection and over-voltage cutoff.
- Integrated reverse-polarity/ESD protection, solar voltage indicators, and TPS63070-based 5 V USB-C output. designed and validated the circuit in KiCad.

Tech: LM393, TPS63070, MOSFETs, USB-C, KiCad, Power Electronics

Flow Driver – Battery Energy Management System | Fizix Solar Innovation

2026

- Developed a hardware-based energy management system that monitors battery conditions and selects the appropriate charge/discharge path.
- Implemented comparator-based battery condition detection and relay switching for automatic energy-path control.
- Designed the switching logic to continuously manage multiple battery packs while maintaining protected operating conditions.

Tech: Comparators, Relays, MOSFETs, Battery Management, Power Electronics, KiCad

LoRa-Based Smart Ambulance Traffic Control

2025

- Developed a wireless traffic-signal control system using LoRa and GPS to prioritize ambulances.
- Implemented master-slave communication and real-time signal control using Embedded C.

Tech: Arduino, GPS, LoRa SX1278, Embedded C

SKILLS

Firmware: Embedded C/C++, FreeRTOS/RTOS, ARM, ESP32/ESP32-S3, Arduino, GPIO, ADC, STM32, PWM, Timers, Interrupts, SQL, Python

Protocols & Wireless: UART, I²C, SPI, RS232, RS485, LoRa, GSM, GPS, Zigbee

Power Electronics: Battery management, solar power systems, power conversion, protection circuits, MOSFET & relay switching

Analog & Control: Analog circuits, comparator circuits, sensor interfacing, voltage & current sensing, motor control

PCB & Validation: KiCad, PCB design/assembly/testing, BOM preparation, circuit debugging, hardware validation, soldering

Tools & Other: STM32CubeIDE, Arduino IDE, Proteus, TinkerCAD, VS Code

DECLARATION

I hereby declare that the information provided in this resume is true and correct to the best of my knowledge and belief.